Analyzing Historically-disputed Events in Online Encyclopedias

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Abstract

Wikipedia language editions often narrate the same disputed event differently, yet prior work mostly captures this through edit histories, citation overlap or document-level scores rather than the assertions themselves. We present a pipeline that measures cross-lingual divergence between English and Spanish articles at the level of atomic propositions, using open-weight models of at most 8B parameters. Aligned article sections are decomposed into propositions and matched across languages to quantify omission, and matched pairs are judged by QWEN3-8B for contradiction and for framing in agency, causality and language. Parallel to these metrics, we compute section-level editorial controversy from revision histories and measure its association with cross-lingual divergence metrics. We find QWEN3-8B agrees with human annotations at $\kappa \approx 0.76$ on all four judged dimensions. On seven territorial-dispute article pairs, omission is higher from English to Spanish and language framing scores highest, while controversy shows no detectable association with divergence. Given the small corpus, these results should be read as exploratory; in particular, the null result for controversy may reflect limited statistical power rather than a true absence of effect. We plan to validate the methodology on a larger corpus covering more disputes between Spanish- and English-speaking territories.

1 Introduction

Wikipedia is a main source of information about the past and a major training and retrieval source for language models (Touvron et al., 2023; Lewis et al., 2020). Each language edition is written by its own community, and for events involving a dispute between groups these communities do not always tell the same story: actors, causes and vocabulary can differ. Since such differences may propagate into downstream systems, understanding precisely how editions diverge matters for the study of collective memory and for evaluating AI-generated encyclopedic content.

Prior work has examined edit-history signals of disagreement (Yasseri et al., 2012; Borra et al., 2015; Bykau et al., 2015), corpus-level framing and sourcing across editions (Oeberst et al., 2020; Xia and Gomez-Zara, 2026; Creanga et al., 2026), and geopolitical bias in LLMs (Li et al., 2024a; Kim and Kim, 2025). What remains under-examined is the content of the divergence itself: which assertions one edition makes that another omits, contradicts, or attributes differently. We address this with a pipeline for aligned English–Spanish articles that aligns sections, decomposes them into atomic propositions, derives directional omission scores from cross-lingual matching, and judges matched pairs for contradiction and framing. Section-level editorial controversy is computed in parallel from revision histories. We ask:

- **RQ1.** Can small open-weight models measure narrative divergence and are they aligned with human judgement?
- **RQ2.** How is divergence distributed across omission, contradiction, and framing?
- **RQ3.** Does section-level editorial controversy predict cross-lingual divergence?

Our contributions are: (i) a locally runnable pipeline that separates omission, contradiction and framing while keeping every score traceable to its source text; (ii) empirically supported design choices, namely sentence-level extraction with QWEN3-8B and translation before embedding; (iii) a validation of QWEN3-8B as a judge against human annotations; and (iv) an analysis of seven disputed territorial events. [1]. Given the small corpus, the main contribution is the methodology rather than strong empirical claims.

[1]Code and data: https://github.com/55horner0/historically-disputed-events-online-encyclopedias

2 Related Work

2.1 Editorial Conflict and Structural Controversy Detection

Early studies of controversy on Wikipedia focused on how disagreements appear through the way users edit and interact with articles. Yasseri et al. (2012) formalized editorial conflict via the mutual reversion metric, showing that disputes manifest as consistent reverting of edits and long-lasting disagreements on talk pages. Yasseri et al. (2014) expanded on this crosslingually to reveal that while some controversial topics are shared globally, most are language dependent. Moving to controversy at the topic-level, Borra et al. (2015) introduced *Contropedia* to track controversy measured at specific wikilinks and sections, while Bykau et al. (2015) introduced the Controversy Detection Algorithm (CDA), which tracks repeated changes in text and hyperlinks across revision histories to identify specific controversies within an article. Benslimane et al. (2021) combined user interaction graph neural networks with text representations to identify controversy in online discussions such as Reddit, while mentioning that it could be expanded to online encyclopedias.

2.2 Cross-Lingual Framing and Collective Memory Bias

Beyond metadata, independent language editions diverge significantly in narrating inter-group conflict. Oeberst et al. (2020) found that communities consistently describe their own group as more moral and less culpable. Analyzing 2,542 battles in 20 languages, Xia and Gomez-Zara (2026) used connotation frames to measure power, agency, and sentiment, and found significant differences in the narrative when language communities were involved, while uninvolved communities tended to be more similar. Creanga et al. (2026) study three contested events in Romanian history across five editions (Romanian, Hungarian, Russian, Turkish, and English). They report "citation isolation", with only 2 of 119 citations shared across three editions of the Battle of Posada article, and track LLM-assigned narrative scores across article snapshots from 2005 to 2024. At the sentence level, Aleksandrova and Lareau (2019) developed multilingual classifiers to detect Neutral Point of View (NPOV) violations, while Borkakoty and Espinosa-Anke (2024) benchmarked automated methods for distinguishing hoaxes from legitimate encyclopedic articles. He et al. (2018) showed that Wikipedia language editions often use substantially different images to represent the same concepts, which may partly reflect cultural contextualization.

2.3 Geopolitical Bias in LLMs and AI Encyclopedias

As large language models (LLMs) ingest encyclopedic text, cross-lingual divergences propagate into generative outputs. Li et al. (2024a) curated *BorderLines*, showing that multilingual LLMs systematically align territorial sovereignty answers with the prompt language's geopolitical stance. In retrieval-augmented generation (RAG), Li et al. (2024b) found that multilingual Wikipedia contexts shift the locus of bias to the retrieved passages, where cross-lingual claims often conflict. Focusing on East Asian disputes, Kim and Kim (2025) further distinguished between intrinsic training-set model bias from prompt-language inference bias. Extending beyond human consensus, Eibl et al. (2026) audited the LLM-generated *Grokipedia*, finding increasing semantic divergence from Wikipedia across article sections and a greater tendency to position right-leaning content near the top of controversial articles. Creanga et al. (2026) additionally compare prompting strategies for generating summaries that attribute, rather than resolve, conflicting claims, evaluating them with an LLM judge. Finally, historical revisionism audits (Ortu et al., 2026) show that LLMs are largely factual under neutral prompting but readily shift to revisionist narratives when explicitly requested through omission, sanitization, and false balance.

2.4 Research Gap and Project Positioning

Prior work thus characterizes disputed narratives from three complementary angles: edit-history signals showing where editors disagree (Yasseri et al., 2012; Borra et al., 2015; Bykau et al., 2015), corpus-level measures of framing, stance, and sourcing across editions (Oeberst et al., 2020; Xia and Gomez-Zara, 2026; Creanga et al., 2026), and benchmarks of LLM geopolitical bias built largely on short or synthetic prompts (Li et al., 2024a; Kim and Kim, 2025). What remains under-examined is the content of the divergence itself: which specific assertions one edition makes that another omits, contradicts, or attributes differently. Creanga et al. (2026) come closest, extracting and matching conflicting claims within a pipeline for generating attributed summaries. However, their analysis of di-

vergence rests on citation stance and LLM-assigned narrative scores, they report no claim-level breakdown of divergence types, and they do not relate divergence to editorial controversy.

Our project addresses this gap through three research questions: **(i)** whether small open-weight models can reliably measure narrative divergence in agreement with human judgments; **(ii)** how divergence is distributed across omission, factual contradiction, and framing differences in agency, causality, and language; and **(iii)** whether section-level editorial controversy predicts cross-lingual divergence. We investigate these questions through aligned English–Spanish Wikipedia articles on historically disputed territorial events.

3 Methodology

Our aim is to characterise *what* diverges when two language editions describe the same historically disputed event, and to relate that divergence to editorial conflict. We study seven such events, selected among the controversial territorial-dispute articles of Wikipedia, and retrieved on October 5, 2026: *Falkland Islands*, *Status of Gibraltar*, *Serranilla Bank*, *Bajo Nuevo Bank*, *Guantanamo Bay Naval Base*, *San Andrés and Providencia* and *Geography of Puerto Rico*; the first pair also served as the pilot. For each event, the English (EN) and Spanish (ES) articles form a pair, and the pipeline aligns their sections (§3.1), decomposes every section into atomic propositions (§3.2) and computes four section-level metrics (§3.3): controversy from the revision histories, omission from a cross-lingual matching of the propositions, and contradiction and framing from an LLM judgement of the matched pairs. The judge and its validation against human annotations are described in §4.5.1. The seven pairs are processed with identical settings, and their outputs are pooled for the joint analysis (§3.4). All components are open-weight models of at most 8B parameters, so the whole pipeline runs on a single machine. Figure 1 summarizes the pipeline.

3.1 Cross-Lingual Section Alignment

Sections are delimited by level-2 headings using the HTML output (subsections are merged into their parent) and extracted independently, giving $\{S_A^i\}_{i=1}^m$ and $\{S_B^j\}_{j=1}^n$. Named entities are linked to Wikidata (Vrandečić and Krötzsch, 2014) from the article's wikilinks (prose and infobox), extended to their surface forms and redirects; e.g. *Argentina* is tagged `Q414`. These QIDs are used by the controversy metric.

Because language editions organise the same event under different headings, sections cannot be matched by title. Instead, each Spanish section is first translated into English with OPUS-MT (Tiedemann and Thottingal, 2020), so that both editions are compared in the same language. Every sentence of one section is then scored against every sentence of the other with BERTScore F1 (Zhang et al., 2020), computed with RoBERTa-base (Liu et al., 2019); each sentence keeps its best match, and the similarity $\mathrm{sim}(S_{\mathrm{EN}}^i, S_{\mathrm{ES}}^j)$ of the two sections is the median of these best-match scores taken in both directions. The Hungarian algorithm then selects the one-to-one assignment that maximises the total similarity,

$$\sigma = \arg\max_{\sigma'} \sum_i \mathrm{sim}\big(S_{\mathrm{EN}}^i, S_{\mathrm{ES}}^{\sigma'(i)}\big), \qquad (1)$$

and only the pairs with $\mathrm{sim} \geq 0.80$ are kept. The resulting aligned pairs $\big(S_{\mathrm{EN}}^i, S_{\mathrm{ES}}^{\sigma(i)}\big)$ localise divergence within the narrative; for readability we write S_{ES}^j for $S_{\mathrm{ES}}^{\sigma(i)}$ below.

3.2 Atomic Proposition Extraction

Each section is decomposed into *atomic propositions* (APs), that is, minimal, self-contained factual statements that can be compared independently across editions. Every sentence of the section is sent separately to QWEN3-8B (Yang et al., 2025) with a prompt, written in the language of the article, that states eight rules, from atomicity and resolved pronouns to faithfulness and no external knowledge. Decoding is greedy and constrained by a JSON schema to a list of strings, so every call returns a parseable output. Each AP is stored as a JSON record with its source sentence, section, article and language, so that matched and omitted propositions can be traced back to the text.

Choice of model. We compared QWEN3-8B with the propositioner used in the pilot, MPROPOSITIONNEUR-V2 (Pommeret et al., 2026), a 0.6B multilingual model distilled from Qwen3-32B, on 300 PropSegmEnt sentences (Chen et al., 2023) and their Spanish translation (Table 1). QWEN3-8B obtains the higher F1 in both languages (0.84 vs. 0.78 in English, 0.75 vs. 0.68 in Spanish) with the same recall, so the gain comes almost entirely from precision (+0.11 and +0.15). MPROPOSITIONNEUR-V2 over-generates (836 and 852 propositions for 784

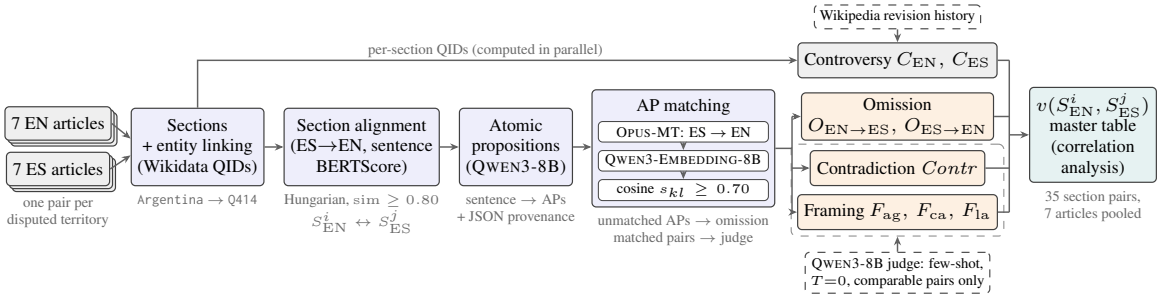


Figure 1: Overview of the final pipeline, run with identical settings on each of the seven EN–ES article pairs. Spanish sections are translated into English and aligned one-to-one with their English counterparts (sentence-level BERTScore, Hungarian matching), and every section is decomposed into atomic propositions (APs) with QWEN3-8B. Spanish APs are translated into English, embedded with QWEN3-EMBEDDING-8B and matched to the English APs by cosine similarity: unmatched APs give the two directional omission scores, and matched pairs are judged by QWEN3-8B for contradiction and framing. Controversy is computed in parallel from the revision history, and the four metrics form the vector v of each aligned section pair; the vectors of the seven articles are pooled in the master table.

	English		Spanish	
	MProp.	Qwen3	MProp.	Qwen3
Precision	0.76	**0.87**	0.65	**0.80**
Recall	0.81	0.81	0.71	0.71
F1	0.78	**0.84**	0.68	**0.75**
Predicted APs	836	728	852	700
Invalid outputs	4	0	5	0

Table 1: Atomic proposition extraction with MPROPOSITIONNEUR-V2 (MProp.) and QWEN3-8B (Qwen3) on 300 PropSegmEnt sentences per language (784 reference propositions each). Predictions are paired with references by the Hungarian algorithm and are correct at a QWEN3-EMBEDDING-8B cosine ≥ 0.80; scores are micro-averaged. Invalid: empty or unparseable output.

references), with repeated or invented statements and returns no usable output for 4 English and 5 Spanish sentences, whereas constrained decoding always gives QWEN3-8B a valid list.

3.3 Section-Level Metrics

Four metrics are computed for every aligned section pair. Controversy is an *intra-article* signal derived from edit histories; omission, contradiction and framing are *cross-article* measures of divergence computed on the APs of the two sections: omission counts the propositions that find no counterpart in the other edition, while contradiction and framing compare those that do. Controversy and omission are computed deterministically, whereas contradiction and framing are evaluated using QWEN3-8B as a judge.

3.3.1 Controversy

Controversy is computed per edition from the article's own revision history, following Borra et al. (2015). Consecutive edits by the same user are collapsed, and vandalism and its reverts (known Wikipedia anti-vandal bots, anonymous edits reverted within 60 s, hash-identical restorations) are discarded. Each revision R_n is compared sentence by sentence with R_{n-1}; $D(R_n)$ is the set of sentences deleted or modified, while pure insertions and new sections are ignored. The controversy of entity q is

$$C(q) = \sum_{n=1}^{N} \sum_{\substack{x \in D(R_n) \\ q \in x}} \frac{1}{w(x)}, \qquad (2)$$

where $w(x)$ is the number of distinct QIDs linked in x (old and new links for a modified sentence). This is the same formula used by Contropedia (Borra et al., 2015).

$C(q)$ is then distributed over the sections of the current version in proportion to the mentions of q in each section, and normalised by section length in words. QIDs absent from the current version are not distributed. Each aligned pair thus carries the controversy in Spanish and English, $\tilde{C}_{\mathrm{EN}}$ and $\tilde{C}_{\mathrm{ES}}$.

3.3.2 Proposition Matching and Omission

Omission measures which propositions of a section have no counterpart in its aligned section. Let $P_{\mathrm{EN}} = \{a_k\}_{k=1}^{K}$ and $P_{\mathrm{ES}} = \{b_l\}_{l=1}^{L}$ be the APs of S^i_{EN} and S^j_{ES} (§3.2). Counterparts are found in four steps (Figure 3):

1. **Translation.** Spanish APs are translated into English with OPUS-MT, $b_l \mapsto \tilde{b}_l$, so both sets are compared in one language.
2. **Embedding.** Each AP is encoded by QWEN3-EMBEDDING-8B (Zhang et al., 2025) as a semantic vector. We use the hidden state of the final token as the sequence representation.
3. **Cosine similarity.** $s_{kl} = \mathbf{e}(a_k)^\top \mathbf{e}(\tilde{b}_l)$ for every cross-lingual pair.
4. **Matching.** A pair is matched when $s_{kl} \geq \tau$; matching is many-to-many, since one edition may state in one proposition what the other splits into several.

An AP without any match is considered omitted in the other edition. As omission is directional, two scores are computed:

$$O_{\text{EN}\to\text{ES}} = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}\Big[\max_{l}\, s_{kl} < \tau\Big], \qquad (3)$$

and symmetrically $O_{\text{ES}\to\text{EN}}$ over the L Spanish APs: $O_{\text{EN}\to\text{ES}}$ is the proportion of English propositions with no counterpart in the Spanish section, and $O_{\text{ES}\to\text{EN}}$ the converse. When one section has no AP, every AP of the other counts as omitted. The matched pairs of both directions, $M = \{(a_k, b_l) : s_{kl} \geq \tau\}$, go on to the contradiction and framing metrics with their original Spanish wording, so the translation is only used to find counterparts.

Threshold evaluation. We studied τ on the Falklands pilot: 58 equivalent EN–ES proposition pairs were annotated by hand in five manually aligned section pairs (43,462 candidate pairs), and τ was swept from 0 to 1 in steps of 0.001. When the multilingual embeddings of the English and the original Spanish APs were compared directly, the best F1 was only 0.53 ($\tau_{F1} = 0.80$; precision 0.69, recall 0.43): more than half of the annotated equivalences were missed, which biases omission upwards. Since a missed equivalence becomes a spurious omission, whereas a false match can at most hide a real one, we instead selected τ by maximising F1.5, which weights recall 1.5 times as much as precision. The best F1.5 was 0.55 ($\tau_{F1.5} = 0.70$; precision 0.36, recall 0.72), recovering 42 of the 58 equivalences. Recall remains limited by a ceiling in the cross-lingual similarity scale: even a sentence and its own Spanish translation reach a similarity of only about 0.88, which leaves a margin of just 0.18 above $\tau_{F1.5}$ for genuine paraphrases. Translating the Spanish APs first removes this ceiling, since both sides are then compared in the same language and identical statements score 1. The final pipeline therefore translates before embedding and uses $\tau_{F1.5} = 0.70$ for all seven article pairs, leaving a margin of about 0.30 for paraphrases.

3.3.3 Contradiction

Contradiction is evaluated on the matched proposition pairs using QWEN3-8B. First, the model determines whether the two propositions are *comparable*, meaning that they refer to the same specific fact, event, claim, or status. Only comparable pairs, denoted by $M_c \subseteq M$, are included in the contradiction analysis.

For each comparable pair, QWEN3-8B then determines whether the two propositions make mutually incompatible factual claims, such that both cannot be true under the same interpretation. Differences in emphasis, perspective, or level of detail are not considered contradictions. The labels YES, UNCLEAR, and NO are assigned values of $c_p = 1$, 0.5, and 0, respectively. The section-level contradiction score is computed as

$$Contr = \frac{1}{|M_c|}\sum_{p\in M_c} c_p. \qquad (4)$$

The comparable pairs labelled NO form the set of comparable non-contradictory pairs $M_{c,nc} = \{p \in M_c : c_p = 0\}$, so that $M_{c,nc} \subseteq M_c \subseteq M$.

3.3.4 Framing

Framing differences are analysed on the comparable non-contradictory pairs, where both editions make compatible claims about the same content but present it differently. Each pair $p \in M_{c,nc}$ is scored by the judge along three dimensions: **agency** (who is portrayed as acting, initiating or being responsible), **causality** (how causes, consequences and motivations are attributed) and **language** (lexical choice, tone, intensity and sentiment). Each dimension receives an ordinal label of SAME OR EQUIVALENT, SLIGHTLY DIFFERENT, or VERY DIFFERENT, encoded as 0, 1, and 2. Averaging over the non-contradictory comparable pairs of a section yields a section-level framing profile,

$$F_d = \frac{1}{|M_{c,nc}|}\sum_{p\in M_{c,nc}} f_d^{(p)}, \quad d \in \{\text{ag}, \text{ca}, \text{la}\}. \qquad (5)$$

3.4 Master Table and Global Analysis

All section-level outputs of the seven article pairs are consolidated into a master table with one row per aligned section pair. Alignment yields 35 section pairs, each containing at least one comparable matched pair, shown in Appendix A (Table 6). Besides the number of APs of each section and of matched pairs, each row stores the vector

$$v\big(S^i_{\mathrm{EN}}, S^j_{\mathrm{ES}}\big) = \big(\tilde{C}_{\mathrm{EN}}, \tilde{C}_{\mathrm{ES}}, O_{\mathrm{EN}\to\mathrm{ES}}, O_{\mathrm{ES}\to\mathrm{EN}}, Contr, F_{\mathrm{ag}}, F_{\mathrm{ca}}, F_{\mathrm{la}}\big). \tag{6}$$

The final analysis examines the relations among the dimensions of the vector across these 35 section pairs. It tests the hypothesis that sections with higher editorial controversy, measured as the mean of $\tilde{C}_{\mathrm{EN}}$ and $\tilde{C}_{\mathrm{ES}}$, show higher omission, contradiction and framing scores (§5.3). This in turn indicates whether historically disputed topics systematically produce divergent representations across language editions.

4 Experimental Setup and Pilot Case

For this pilot study, we focused on the English and Spanish versions of the Wikipedia article on the *Falkland Islands* (Q9648), titled *Islas Malvinas (territorio británico de ultramar)* in Spanish. This article pair was used as the initial case to develop, test, and refine the different components of the pipeline. Once the pipeline described in Section 3 had been developed through this pilot study, it was scaled to the six remaining article pairs included in the study. All stages use open-weight models and run locally on a single machine: `opus-mt-es-en`, a compact OPUS-MT translation model, translates Spanish sections and propositions into English; RoBERTa-base computes BERTScore for section alignment; QWEN3-8B, runed with `llama.cpp` (Gerganov and contributors, 2023), extracts the atomic propositions and acts as the contradiction and framing judge; and QWEN3-EMBEDDING-8B embeds the propositions for matching. Table 2 summarises the data kept at each stage.

4.1 QID extraction and entity linking

We considered several approaches from simply obtaining QIDs present in the Wikipedia article, which resulted in low recall since many entities are linked only once but appear multiple times, to using Wikidata aliases table and anchor text, which reduced precision considerably ("per" was identified as Peru for example). Finally, we decided to limit QID extraction to QIDs present in the text and info box, as well as their corresponding aliases, which would get tagged with their respectives QIDs.

Article pair	APs EN/ES	Aligned sections	Matched pairs	Compar. pairs
Falklands	426/271	7	139	130
Bajo Nuevo	112/45	3	69	62
Puerto Rico	290/247	5	263	217
Guantanamo	418/152	5	174	166
San Andres	199/236	5	82	71
Serranilla	133/80	3	70	65
Gibraltar	554/394	7	573	473
Total	**2132/1425**	**35**	**1370**	**1184**

Table 2: Data kept at each stage for the seven article pairs: atomic propositions per edition, aligned section pairs ($\mathrm{sim} \geq 0.80$), matched AP pairs ($s_{kl} \geq 0.70$) and matched pairs judged comparable by QWEN3-8B.

4.2 Section Alignment

The alignment takes as input the sections obtained directly from the output of our Wikipedia QID extraction pipeline, and it went through several iterations before it worked as expected. Our first approach compared the full text of each section of article A with the full text of each section of article B using a multilingual BERTScore F1. This did not work: the F1 always fell between 0.55 and 0.70, so no pair stood out as a clear match. To obtain clearer links between sections, we tried several chunking strategies, from character limits to the first sentence of each section and the first sentence of each paragraph, but none of them improved the scores.

To understand why, we computed the BERTScore between two sentences, one in Spanish and one in English, that were direct translations of each other and should therefore have been highly similar. Even in this case the score was quite low, which was our sign that we needed to change our method.

We therefore translated the Spanish sections into English and computed the BERTScore between the two English texts. As we were concerned that translation might introduce biases, the translated text was used only for this similarity computation: the rest of the work relies on the original texts, and we translate only when needed for text similarity.

To further refine the alignment, we implemented a best-sentence median approach (§3.1). Rather than comparing entire sections as single blocks of

text, we split each section into sentences and computed the BERTScore between the sentences of the two sections. For each sentence of one section, we kept its highest score against any sentence of the other, and we aggregated these best-match scores with the median, repeating the process in both directions. This measured how consistently the sentences of each section had a semantically similar counterpart in the other, instead of letting a few highly similar sentences dominate the overall score. The aggregated scores formed a section-pair similarity matrix, from which the Hungarian algorithm selected a one-to-one alignment between sections, and a minimum similarity threshold finally discarded the pairs whose scores were too low to be considered reliable matches.

We also checked the resulting matches with our own judgement to make sure they made sense. Once the method was confirmed on the pilot case, we applied it to all articles, investigating only the matches that seemed suspicious.

4.3 Atomic Proposition Extraction

Like the section alignment, the extractor went through four versions (E1–E4, Figure 2).

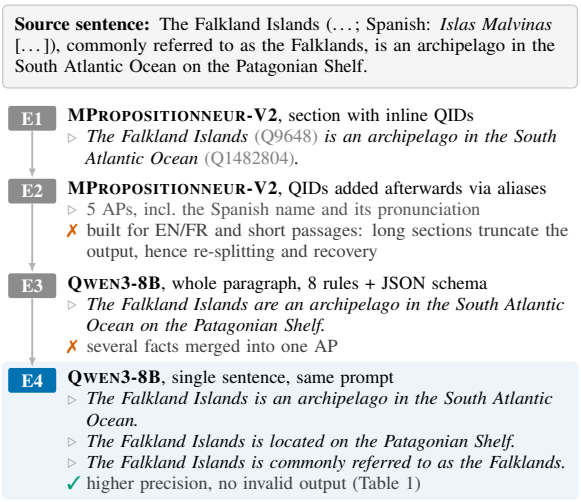


Figure 2: Versions of the AP extractor (E1–E4) and their verbatim output for the first sentence of the English Falklands article. Superseded versions are in grey, each with the problem that led to the next one.

E1–E2: MPROPOSITIONNEUR-V2. This model receives a passage after the instruction *Atomize:* and returns a JSON list of propositions. E1 wrote the QID of every linked entity next to its mention, keeping the identifiers needed for the triplets of §4.4; E2 removed them from the input and attached them to the triplets afterwards, by matching entity labels against the aliases of the article. The model was designed mainly for English and French and for short passages, due to that the model needed fallbacks on our long sections: splitting a chunk again when its output was cut off, and recovering the complete propositions from truncated outputs.

E3–E4: QWEN3-8B. The final extractor uses this model to generate propositions in the language of the article, following eight rules (atomicity, self-contained propositions, faithfulness to the source, factual precision, minimality, no duplicates, no external knowledge and no stronger reformulations), and constrains decoding with a JSON schema so that every call returns a valid list. Run on whole paragraphs (E3), which in our extracted text amounted to entire sections, it merged several facts into single propositions (32 for the lead section, against 53 at the sentence level), so we moved it to the sentence level (E4), where it is also more precise than MPROPOSITIONNEUR-V2 (§3.2, Table 1).

4.4 Omission Extraction and Equivalent Atomic Propositions

The way we measured omission changed as much as the extractor: it moved from entities to claims, and then to claims compared in a single language (O1–O4, Figure 3).

O1: entities. From every proposition with at least two QIDs, an LLM extracted triplets $t = (q_s, r, q_o)$ linking two of its QIDs through a relation r, and omission was computed as in Eq. 3 over the QIDs of the triplets: the share of the QIDs of one section that do not appear in the triplets of the other. This measured which entities were missing rather than which claims.

O2: verbalised triplets. Each triplet was verbalised as "subject relation object", embedded with QWEN3-EMBEDDING-8B and counted as omitted when no triplet of the other section reached a cosine similarity of 0.70. Reducing a statement to three slots, however, easily drops dates, figures and qualifiers (Figure 3a).

O3: propositions. We therefore matched the APs themselves, keeping every counterpart above the threshold rather than only the best one, since one edition may state in one proposition what the other

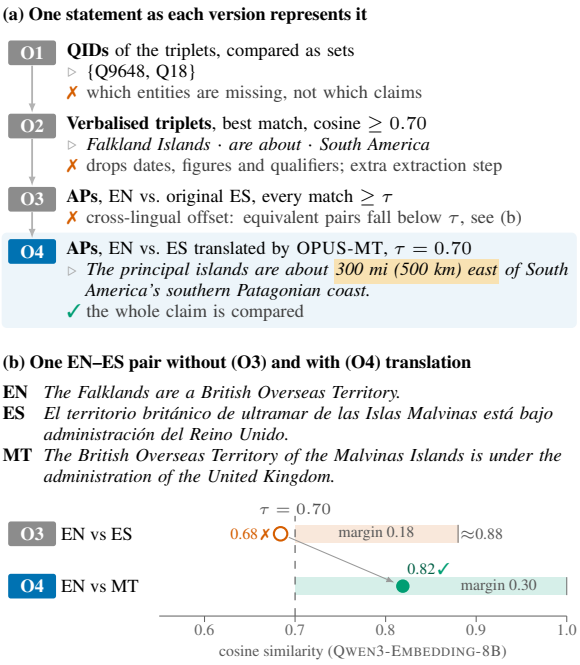


Figure 3: Versions of the omission measure (O1–O4), on real pipeline outputs. (a) One English statement as each version represents it. (b) A pair judged comparable, with the Spanish AP embedded directly (O3) or after translation (MT, O4).

splits into several. However, when compared directly with the original Spanish APs, even a sentence and its own translation achieve a similarity of only about 0.88 (§3.3.2). This leaves little room above the threshold, causing some genuinely equivalent pairs to fall below it (Figure 3b).

O4: propositions in one language. The final version translates the Spanish APs into English with OPUS-MT before embedding them, so that both sides are compared in one language, and applies $\tau = 0.70$. The pairs above the threshold are the equivalent APs, which are sent to the LLM judge with their original Spanish wording.

4.5 Model-based Metrics

4.5.1 Human Judgment

To obtain a human reference for the evaluation, we randomly extracted a total of 100 pairs of atomic propositions (Spanish–English) from the different articles and organised them into a spreadsheet. Each of the 100 proposition pairs was manually annotated along four dimensions: **contradiction**, **agency framing**, **causality framing**, and **language framing**. Some annotations were carefully reviewed and cross-checked with state-of-the-art LLMs (Claude Opus 4.6 (Anthropic, 2026) and GPT-6.1 Sol (OpenAI, 2026)) for additional verification (rationale behind it), before being compared with the model's predictions.

Among the 91 pairs for which a contradiction judgement was available, 37 were labelled contradictory, corresponding to a contradiction rate of $37/91 = 40.66\%$, and one was unclear. Framing was evaluated on the 53 eligible non-contradictory pairs. Language showed the strongest framing difference, with an average score of $78/53 = 1.47$, followed by causality at $72/53 = 1.36$ and agency at $64/53 = 1.21$. These results (Table 3) suggest that, even when the two editions express compatible claims, they frequently differ in wording, tone, emphasis, agency, or causal interpretation.

Metric	Score
Contradiction	40.66%
Agency Framing	1.21
Causality Framing	1.36
Language Framing	1.47

Table 3: Human annotation results for contradiction and framing over the sampled proposition pairs.

4.5.2 Model Prompting Strategy

We progressively refined the prompting strategy used by QWEN3-8B to evaluate contradiction and framing differences between matched atomic propositions. Figure 4 illustrates the four iterations. Initially, standard prompting (P1) provided only the proposition pair and a classification question. Few-shot examples were then introduced (P2), but without explicit definitions, the distinction between evaluation categories remained ambiguous. In P3, detailed task descriptions were added alongside the examples. Finally, P4 refined these descriptions using reasoning outputs from Claude Opus 4.6 to clarify the evaluation criteria.

The final configuration uses 12 annotated samples as few-shot examples for each metric, combined with the refined task descriptions. We also tested different prompt formulations and decoding parameters, observing better results when the English proposition preceded the Spanish one. This configuration was retained for the final evaluation.

5 Results and Analysis

5.1 Model–Human Agreement

To evaluate the reliability of QWEN3-8B as an LLM judge, we compared its predictions against human annotations on the 100 randomly selected

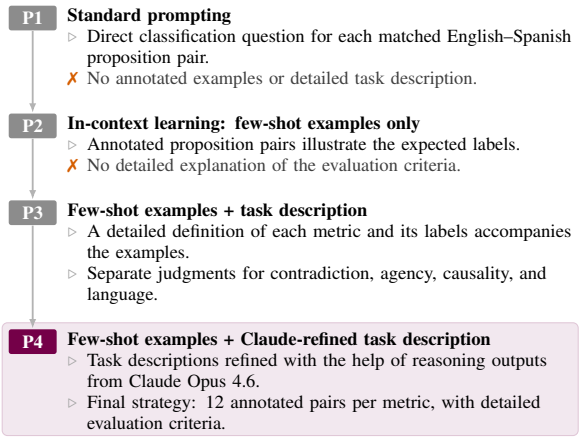


Figure 4: Evolution of the prompting strategy (P1–P4), from standard prompting to few-shot learning with Claude-assisted task descriptions.

English–Spanish atomic proposition pairs. Model performance was assessed using raw agreement, Cohen's κ (Cohen, 1960), which corrects agreement for chance, and macro-averaged F1.

Metric	Agreement	Cohen's κ	Macro F1
Comparability	0.988	0.794	0.897
Contradiction	0.888	0.761	0.880
Agency	0.883	0.759	0.879
Causality	0.863	0.760	0.848
Language	0.883	0.760	0.848

Table 4: Agreement between QWEN3-8B predictions and human annotations for contradiction and framings.

As shown in Table 4, QWEN3-8B achieves consistently strong agreement across all four dimensions, with agreement scores ranging from 0.863 to 0.888 and Cohen's κ values around 0.76. Contradiction achieves the highest agreement (0.888), while causality is slightly more challenging (0.863). These results indicate substantial agreement beyond chance, supporting the use of QWEN3-8B for evaluating cross-lingual narrative divergence. Nevertheless, the agreement is strong but not perfect, suggesting that QWEN3-8B captures our judgment patterns without simply copying them. This is desirable because we do not treat the human annotations as an absolute ground truth: in a subjective and complex evaluation setting like ours, human judgments can also vary or contain errors.

5.2 Typology of Divergence

Across the 35 aligned section pairs (Appendix A), divergence combines asymmetric coverage, factual disagreement, and differences in presentation. The

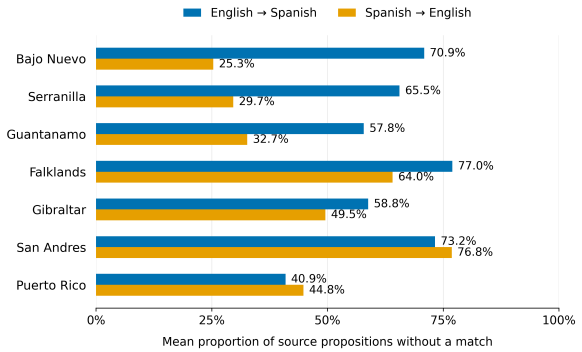


Figure 5: Mean directional omission by article pair. Percentages are shown beside the bars; article means give equal weight to aligned section pairs.

patterns vary across articles, while language framing consistently receives the highest scores among the framing dimensions.

Omission is asymmetric, but its direction varies. Mean omission is higher from English to Spanish (0.634) than from Spanish to English (0.495), a gap of 13.9 percentage points. EN-to-ES omission is higher in 22 of the 35 section pairs and in five of the seven article-level means.

Figure 5 shows that this asymmetry is particularly pronounced for Bajo Nuevo and Serranilla, where the article-level gaps reach 45.6 and 35.9 percentage points, respectively. Guantanamo also shows a substantial gap of 25.2 points. Puerto Rico and San Andres exhibit the reverse pattern, with ES-to-EN omission exceeding EN-to-ES omission by 3.9 and 3.6 points. These reversals show that the balance of coverage depends on the article. Ten section pairs also have omission scores of at least 0.80 in both directions, revealing substantial two-way differences in the propositions covered by the aligned sections.

Factual disagreement and language framing recur across articles. The contradiction score has a mean of 0.367 and a median of 0.375, with nonzero scores in 34 of the 35 section pairs (Figure 6). Article-level means are highest for Bajo Nuevo (0.460), followed by the Falklands (0.446) and Gibraltar (0.421), and lowest for San Andres (0.266) and Puerto Rico (0.269). Factual disagreement therefore appears across the corpus, with different intensities between articles.

Language framing has the highest mean (1.064), followed by causality (0.708) and agency (0.625). This ordering is especially consistent for language: it has the highest article-level mean in all seven

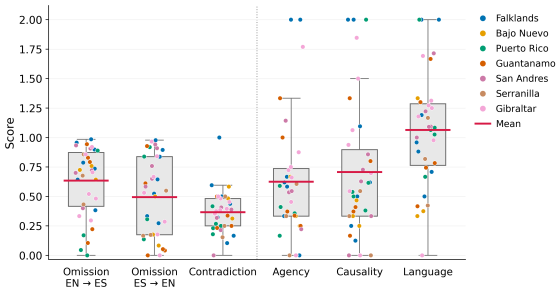


Figure 6: Section-level divergence scores across all 35 aligned section pairs. Coloured points identify articles, grey boxes show interquartile ranges, whiskers extend to observations within 1.5 interquartile ranges, and red bars indicate means.

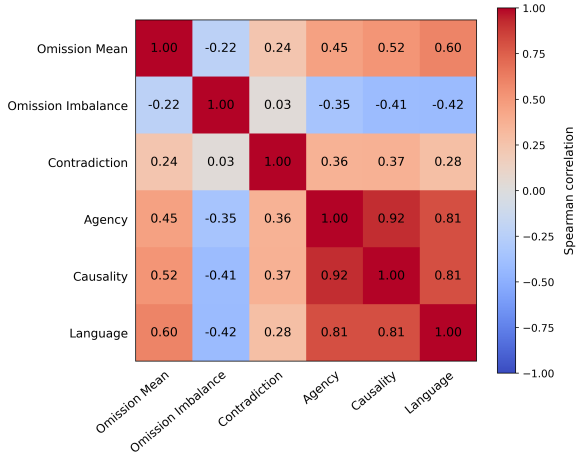


Figure 7: Spearman rank correlations across all 35 section pairs. Cell annotations report coefficients rounded to two decimals; colours use a fixed scale from -1 to $+1$.

articles and is highest or tied for highest in 33 of the 35 section pairs. Differences in wording, tone and intensity are consequently a recurring feature of the cross-lingual divergence.

The article profiles also reveal different combinations of divergence. San Andres, for example, combines high omission in both directions with the lowest article-level contradiction mean (0.266) and the highest language framing mean (1.224). Its profile illustrates how substantial differences in coverage and presentation can accompany relatively lower factual disagreement.

Framing dimensions co-vary more strongly than contradiction. The correlations in Figure 7 show a strong association between agency and causality ($\rho = 0.92$). Language also correlates strongly with both dimensions ($\rho \approx 0.81$). Sections with greater differences in the portrayal of actors therefore tend to show greater differences in causal attribution and language as well.

Mean omission is positively associated with agency ($\rho = 0.45$), causality ($\rho = 0.52$), and language ($\rho = 0.60$). Coverage differences thus tend to increase alongside framing differences, with the strongest relationship involving language. Contradiction has weaker associations with agency ($\rho = 0.36$), causality ($\rho = 0.37$), and language ($\rho = 0.28$), indicating that factual disagreement follows a less closely shared pattern than the three framing dimensions.

Omission imbalance is negatively associated with agency ($\rho = -0.35$), causality ($\rho = -0.41$), and language ($\rho = -0.42$). Higher framing differences tend to occur in sections with a smaller signed EN-to-ES omission gap. The near-zero association between omission imbalance and contradiction ($\rho = 0.03$) further separates directional coverage asymmetry from the pattern of factual disagreement.

Taken together, the results show an overall EN-to-ES omission asymmetry, recurring factual disagreement, and consistently higher language framing scores than agency or causality scores. Coverage and framing differences are positively related, while the three framing dimensions form the most strongly interconnected part of the divergence profile. The article-level variation shows how these forms of divergence combine differently across the seven cases.

5.3 Controversy and Editorial Link

For each outcome we fit a line $y = \alpha + \beta z$ over the 35 section pairs, where z is the mean of $\tilde{C}_{\text{EN}}$ and $\tilde{C}_{\text{ES}}$ and both y and z are standardised, so that β equals their correlation. Standard errors are clustered by article, with a t-test on 6 degrees of freedom. No coefficient differs significantly from zero at the 5% level (Table 5): section-level controversy does not predict cross-lingual divergence in our sample. Because clustered standard errors can be unreliable with only seven clusters, we also ran a wild cluster bootstrap (Cameron et al., 2008), which keeps the data fixed and randomly reweights the residuals of each article under the null hypothesis. With Webb weights (Webb, 2013) and 9,999 replications, it gives the same conclusion (all $p \geq 0.17$).

Outcome	β Coef.	95% CI	*t-test p*
$\text{Omission}_{\text{EN}\to\text{ES}}$	$+0.15$	$[-0.15, +0.44]$	0.26
$\text{Omission}_{\text{ES}\to\text{EN}}$	$+0.20$	$[-0.18, +0.58]$	0.25
Contradiction	-0.02	$[-0.36, +0.33]$	0.92
Agency	$+0.03$	$[-0.34, +0.40]$	0.85
Causality	$+0.01$	$[-0.36, +0.38]$	0.94
Language	$+0.04$	$[-0.39, +0.47]$	0.83

Table 5: OLS of each divergence outcome on section controversy, both standardised (35 sections, 7 articles, article-clustered SE).

6 Limitations

Inflated controversy scores. The section score assumes that an entity's controversy is distributed proportionally to its mentions, which can inflate scores for short sections containing disputed entities, even if the entity was not controversial in that particular section. English revision histories are generally longer than Spanish ones, so $\tilde{C}_{\text{EN}}$ and $\tilde{C}_{\text{ES}}$ are not necessarily on the same scale. Renamed sections are treated as new, so earlier edits may be ignored, and the alias filters rely on English stopword lists, which can make Spanish entity linking noisier.

Small corpus and limited statistical power. The study is based on seven English–Spanish article pairs, from which 35 aligned section pairs were retained for the final analysis. This sample limits the statistical power of the regression analysis for RQ3, especially because standard errors are clustered by article, leaving only seven independent clusters. The analysis is therefore sensitive mainly to relatively large effects, and the absence of statistically significant associations should not be interpreted as evidence that no relationship exists between editorial controversy and narrative divergence.

Computational and project constraints. The small size of the corpus is largely a consequence of the cost of the pipeline. This cost varies substantially across articles, because inference time depends on article length, on the number of extracted atomic propositions, and on the number of proposition pairs that must be embedded, matched, and evaluated by the LLM. In particular, repeated inference with QWEN3-8B, both to extract propositions and to judge contradiction and framing, made large-scale processing expensive on the available local hardware. Combined with a development period of approximately six weeks, these constraints restricted the final experiment to seven article pairs.

Limited disciplinary and linguistic coverage. The team consisted mainly of computer science students, only two of whom were Spanish speakers. This was sufficient to develop the English–Spanish pipeline and to inspect its outputs manually, but a more interdisciplinary team could provide better-informed interpretations of the narrative differences.

7 Conclusion

We present a pipeline that measures cross-lingual divergence between Wikipedia editions at the level of atomic propositions, using only open-weight models of at most 8B parameters. Applied to seven English–Spanish article pairs on territorial disputes, it answers our three research questions. First, small models are feasible: QWEN3-8B agrees with human annotations for contradiction and for all three framing dimensions. Second, the typology of divergence varies across articles, but omission is consistently higher from English to Spanish, and language framing receives the highest scores: even when two editions make compatible claims, they often present them differently. Third, section-level editorial controversy shows no detectable association with any divergence measure. Given the width of the confidence intervals, this calls for a larger, multilingual corpus rather than the conclusion that no link exists.

8 Discussion

Beyond answering the research questions, the main outcome is the pipeline itself, whose successive versions yielded three lessons that should transfer to other language pairs. The first is to compare texts within a single language: since cross-lingual similarity is compressed into a narrow range, the Spanish side is translated into English to find counterparts, while contradiction and framing are still judged on the original wording. The second is to work at the level of atomic propositions: whole sections or paragraphs blur several facts together, whereas triplet structures lose dates, figures and qualifiers. The third is that small models can be reliable when their task is tightly specified: the extractor relies on explicit rules and schema-constrained decoding, and the judge on explicit criteria for each label and separate judgements per dimension. Every stage that measures divergence was checked against human judgement.

Finally, reporting metrics per aligned section

pair shows where in the narrative between two editions may diverge, and every divergence score can be traced back to the propositions and sentences behind it.

Beyond the study of collective memory, knowing where and how Wikipedia editions diverge also matters for language models. These models are trained on Wikipedia and often retrieve information from it, so the account they give of a disputed event may depend on which edition they draw on.

9 Future work

Future work should address the main limitations identified in this study while improving the robustness, scalability, and generalizability of the pipeline.

Controversy measure. This measure could be complemented with an LLM to determine whether an edit reflects substantive disagreement with the previous version, incorporating talk-page discussions, and tracking sections across revision histories instead of distributing entity-level controversy according to current mentions. This would allow us to keep track of the controversy of entities that disappear in subsequent versions, which could potentially hold significant controversy. Language-specific preprocessing and stopword lists could further improve entity linking in Spanish.

Corpus size. A larger corpus covering additional English–Spanish territorial disputes would provide greater statistical power and support more robust analyses of the relationship between editorial controversy and narrative divergence. Expanding the available computational resources would facilitate this scaling by reducing the constraints imposed by inference time and local hardware.

Annotation. Having a more interdisciplinary team, including specialists in history and politics, as well as additional Spanish-speaking annotators would strengthen the interpretation of historically sensitive cases. Collecting several independent annotations per proposition pair would also make it possible to report inter-annotator agreement. However, with a team of only five annotators, this estimate would remain relatively limited and should be interpreted with caution. A larger annotation pool would provide a more robust basis for human evaluation and for comparing human agreement with that of QWEN3-8B.

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A Section-Level Results

Article	Sec. EN/ES	APs EN/ES	Match. pairs	$O_{\mathrm{EN}\to\mathrm{ES}}$	$O_{\mathrm{ES}\to\mathrm{EN}}$	$Contr$	F_{ag}	F_{ca}	F_{la}	$\tilde{C}_{\mathrm{EN}}$	$\tilde{C}_{\mathrm{ES}}$
Falklands	0/0	53/21	25	0.74	0.52	0.10	0.33	0.25	0.96	0.764	0.329
	1/1	28/11	17	0.64	0.27	0.44	0.00	0.12	0.50	0.416	0.417
	2/2	147/92	22	0.90	0.84	0.60	0.67	1.10	1.19	0.532	0.104
	3/3	61/45	27	0.79	0.64	0.17	0.58	0.50	0.71	0.649	0.108
	6/5	67/19	2	0.99	0.89	0.50	2.00	2.00	2.00	0.313	0.023
	7/6	47/39	45	0.38	0.33	0.32	0.62	0.62	0.88	0.471	0.080
	9/7	23/44	1	0.96	0.98	1.00	2.00	2.00	2.00	0.254	0.081
Bajo Nuevo	0/0	14/16	9	0.64	0.50	0.31	0.25	0.25	0.38	0.184	0.062
	1/1	40/12	31	0.72	0.08	0.48	0.67	0.47	1.33	0.054	0.042
	2/2	58/17	29	0.76	0.18	0.58	0.33	0.33	0.33	0.109	0.042
Puerto Rico	0/1	27/49	4	0.89	0.92	0.25	1.00	2.00	2.00	0.231	0.048
	1/0	47/62	98	0.17	0.31	0.23	0.59	0.54	1.06	0.120	0.089
	2/3	44/44	93	0.05	0.14	0.27	0.36	0.38	0.67	0.054	0.033
	3/4	156/68	16	0.94	0.84	0.42	0.17	0.50	1.08	0.051	0.034
	4/5	16/24	52	0.00	0.04	0.18	0.41	0.62	1.03	0.060	0.052
Guantanamo	0/1	35/55	6	0.83	0.93	0.50	1.33	1.33	1.67	0.380	0.124
	1/2	27/24	44	0.22	0.04	0.23	0.37	0.37	0.74	0.067	0.004
	2/3	178/38	90	0.79	0.05	0.34	0.34	0.41	0.78	0.210	0.012
	3/4	159/18	10	0.94	0.61	0.50	1.00	0.80	1.30	0.145	0.091
	4/5	19/17	24	0.11	0.00	0.25	0.25	0.17	0.42	0.236	0.012
San Andres	0/0	10/20	10	0.40	0.65	0.39	0.22	0.33	1.22	0.204	0.268
	2/1	79/120	51	0.72	0.76	0.41	0.53	0.63	1.09	0.085	0.078
	4/3	75/55	9	0.91	0.91	0.21	1.14	0.86	1.71	0.035	0.091
	5/5	20/24	11	0.70	0.58	0.32	0.55	0.73	1.09	0.053	0.174
	6/6	15/17	1	0.93	0.94	0.00	0.00	0.00	1.00	0.030	0.058
Serranilla	0/0	28/6	6	0.86	0.17	0.25	0.33	0.33	1.17	0.163	0.038
	1/1	37/23	27	0.43	0.17	0.15	0.00	0.00	0.42	0.063	0.007
	2/2	68/51	37	0.68	0.55	0.50	0.61	0.55	0.82	0.092	0.012
Gibraltar	0/0	27/32	34	0.30	0.53	0.45	0.88	0.94	1.31	0.273	0.120
	1/1	48/108	154	0.52	0.67	0.48	0.66	0.54	1.18	0.106	0.030
	2/2	166/105	264	0.48	0.29	0.40	0.72	1.06	1.27	0.060	0.019
	4/3	14/58	4	0.86	0.97	0.38	0.75	1.50	1.25	0.130	0.023
	5/5	122/45	96	0.70	0.18	0.35	0.45	0.70	0.98	0.040	0.014
	6/6	174/44	19	0.92	0.84	0.38	1.77	1.85	1.69	0.071	0.022
	8/10	3/2	2	0.33	0.00	0.50	0.00	0.00	0.00	0.000	0.000

Table 6: Master table: one row per aligned section pair (35 pairs, 7 articles). Sec.: section indices in the EN and ES articles; APs: atomic propositions per section; Match. pairs: matched AP pairs ($s_{kl} \geq 0.70$); $\tilde{C}$: controversy per token.